Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 5: The Macroeconomy in the Medium Run - Deep Crisis

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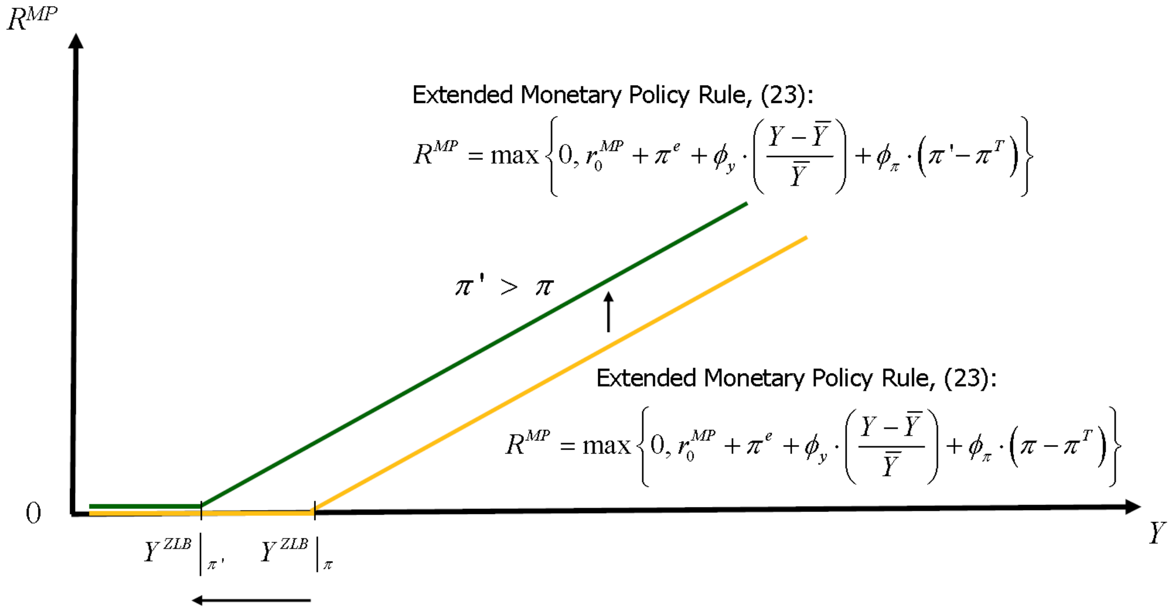
May 12, 2026

+ We return to studying macroeconomic outcomes following the onset of the global financial crisis, combining the medium-run “aggregate supply” relationship (medium-run *AS*-curve) with the insights from the extended *IS-TR* model, to obtain the **extended *AS-AD* model**.
+ In analogy to our modelling strategy for the macroeconomy in the short run, we will augment the financial sector within the *AS-AD* model (that is, the medium-run *TR*-curve) by two features:
  - First feature: a zero lower bound on the nominal monetary policy rate, R^{MP}, which in the medium-run context is captured through the following **extended monetary policy rule**:

$$R^{MP} = \max\left\{0,\ \underbrace{r_0^{MP} + \pi^e + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right)}_{\text{Taylor Rule}}\right\}.$$

  - Under Equation (23), if according to the Taylor rule the central bank would choose to set a negative R^{MP}, it is constrained by the zero lower bound and instead sets $R^{MP} = 0$.
  - When the zero lower bound on R^{MP} is binding, we label the macroeconomy as being in a **”deep crisis”**.


R^{MP}
Extended Monetary Policy Rule, (23):
$R^{MP} = \max\left\{0, r_0^{MP} + \pi^e + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi' - \pi^T\right)\right\}$
$\pi' > \pi$
Extended Monetary Policy Rule, (23):
$R^{MP} = \max\left\{0, r_0^{MP} + \pi^e + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right)\right\}$
0
Y
$Y^{ZLB}\big|_{\pi'}$
$Y^{ZLB}\big|_{\pi}$

- Note from Equation (23) that the highest level of output at which the zero lower bound on R^{MP} is binding in the medium run depends on the rate of inflation, and is thus endogenous. To indicate this dependence, we denote the zero lower bound level of output as $Y^{ZLB}\big|_{\pi}$:

$$r_0^{MP} + \pi^e + \phi_y \cdot \left(\frac{Y^{ZLB}\big|_{\pi} - \bar{Y}}{\bar{Y}} \right) + \phi_\pi \cdot \left(\pi - \pi^T \right) = 0 \tag{24}$$

$$\Leftrightarrow \quad Y^{ZLB}\big|_{\pi} = \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi^e - \phi_\pi \cdot \left(\pi - \pi^T \right) \right] \tag{25}$$

- (25) implies that the zero lower bound level of output depends negatively on the rate of inflation: as the rate of inflation, say, increases, then the central bank sets the monetary policy rate to be higher (as long as the new zero lower bound constraint is not binding), and thus reaches the zero lower bound for R^{MP} at a lower level of output.
- Second feature: an endogenous component to the risk premium, that in ”deep crisis” times drives the mark-up between
  - the interest rate, r, at which households and firms can borrow from the commercial banks to fund (part of) their consumption and investment expenditure, and
  - the real value of the monetary policy rate, $r^{MP} \approx R^{MP} - \pi^e$, set by the central bank:

$$r^{RP} = r - r^{MP} = r_0^{RP} - \underbrace{r_y \cdot \left(Y - Y^{ZLB}\big|_{\pi} \right) \cdot I\left(Y \leq Y^{ZLB}\big|_{\pi} \right)}_{\text{endogenous component of risk premium}} . \tag{26}$$

+ Combining Equations (23) and (26), we obtain the **extended *TR*-curve**, as applicable in the **medium run**:

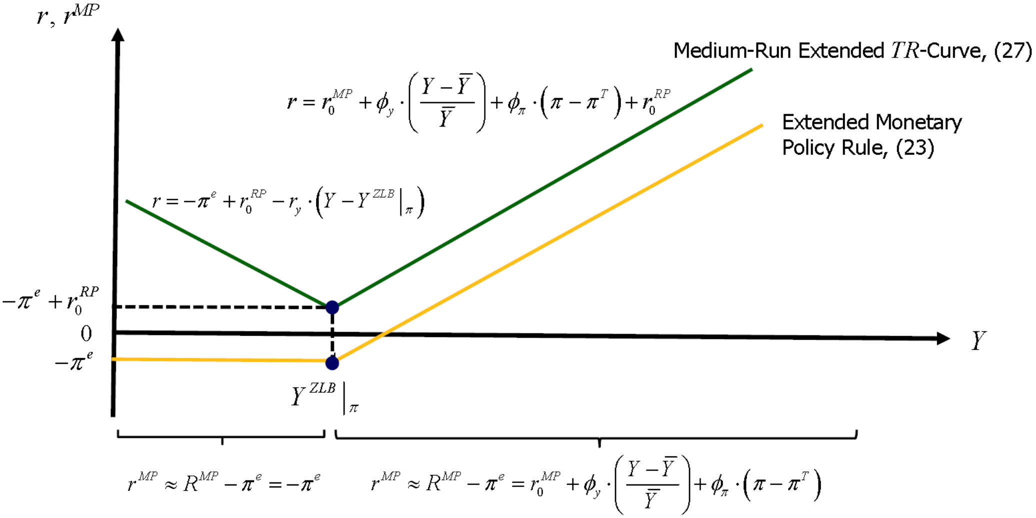
$$\begin{aligned} r &= r^{MP} + r^{RP} \approx R^{MP} - \pi^e + r^{RP} \\ &= \max\left\{-\pi^e,\ r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right)\right\} \\ &\quad + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\big|_\pi\right) \cdot I\left(Y \leq Y^{ZLB}\big|_\pi\right). \end{aligned} \tag{27}$$

+ The medium-run extended *TR*-curve features a discontinuity at that level of output for which the Taylor rule suggests switching from a positive to a zero monetary policy rate, $Y^{ZLB}\big|_\pi$:
+ For "normal business cycle" outcomes, the zero lower bound on R^{MP} is not binding ($R^{MP} > 0$ and $Y > Y^{ZLB}\big|_\pi$), and (27) reduces to the medium-run *TR*-curve given by (11):

$$r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right) + r_0^{RP}.$$

+ For "deep crisis" outcomes, the zero lower bound on R^{MP} is binding ($R^{MP} = 0$ and $Y \leq Y^{ZLB}\big|_\pi$), and (27) reduces to:

$$r = -\pi^e + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\big|_\pi\right). \tag{28}$$


r, r^{MP}
Medium-Run Extended TR-Curve, (27)
$r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right) + r_0^{RP}$
Extended Monetary Policy Rule, (23)
$r = -\pi^e + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\big|_\pi\right)$
$-\pi^e + r_0^{RP}$
0
$-\pi^e$
Y
$Y^{ZLB}\big|_\pi$
$r^{MP} \approx R^{MP} - \pi^e = -\pi^e$
$r^{MP} \approx R^{MP} - \pi^e = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right)$

Workbook for the Medium-Run Extended *TR*-Curve

The workbook *Medium-Run-Extended-TR-Curve.xlsm* allows to study all the determinants of the position and slope of the medium-run extended *TR*-curve.

Medium-Run Extended TR-Curve: Specification

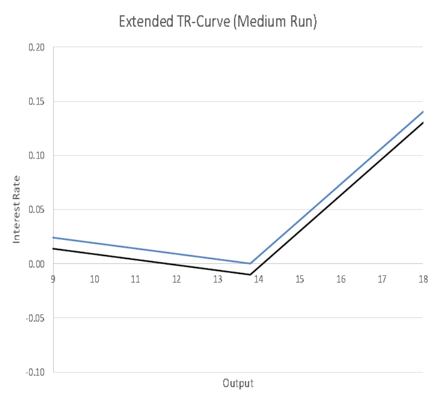
Parameters

	Initial	New
r_0^{MP}	0.020	0.020
r_0^{RP}	0.010	0.020
ϕ_y	0.500	0.500
$\overline{Y}$	15.000	15.000
r_y	0.005	0.005
ϕ_π	2.000	2.000
π^T	0.020	0.020
π	0.020	0.020
π^e	0.020	0.020

Quantitative Results

	Initial	New
$Y^{ZLB}\|_\pi$	13.800	13.800

Medium-Run Extended TR-Curve: Graph



The extended *AS-AD* model combines the medium-run extended *TR*-curve with the standard *IS*-curve, to yield an extended *AD*-curve.

To construct the extended *AD*-curve, let us first consider **"normal business cycle" outcomes**:

+ When the zero lower bound on R^{MP} is not binding, then the extended *AD*-curve is the same as the *AD*-curve in the *AS-AD* model.
+ To see this, note that in the *AS-AD* model the *AD*-curve (14) was obtained from combining the medium-run *TR*-curve (11) that holds when $R^{MP} > 0$ with the *IS*-curve (12), yielding

$$\pi = \frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \cdot Y.$$

+ Thus, when the zero lower bound on R^{MP} is not binding, then the extended *AD*-curve features the familiar negative relationship between output and inflation.
+ This still reflects that as the rate of inflation, say, decreases, then the central bank, following the Taylor rule, lowers its monetary policy rate, R^{MP}. For a given exogenous risk premium, this increases output through the consumption, investment and exchange rate channels.

Turn second to "deep crisis" outcomes:

+ When the zero lower bound on R^{MP} is binding, we can obtain the "deep crisis" portion of the extended *AD*-curve that is applicable then by substituting the real interest rate that holds when $R^{MP} = 0$ (see the medium-run extended *TR*-curve, Equations (27) and (28)),

$$r = -\pi^e + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\big|_\pi\right),$$

into the *IS*-curve (12), obtaining

$$Y = IS_0 - IS_1 \cdot \left[-\pi^e + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\big|_{\pi} \right) \right], \tag{29}$$

where, from Equation (25),

$$Y^{ZLB}\big|_{\pi} = \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi^e - \phi_\pi \cdot \left(\pi - \pi^T \right) \right].$$

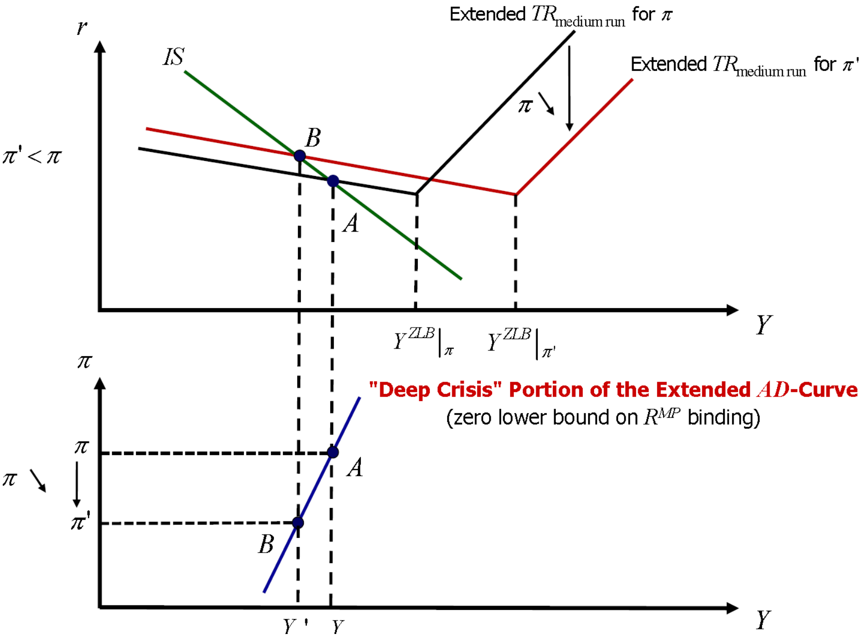
+ To complete derivation of the "deep crisis" portion of the extended *AD*-curve, we need to solve (29) for the rate of inflation. Substituting from (25) back into (29), we obtain

$$\begin{aligned}
Y &= IS_0 - IS_1 \cdot \left[-\pi^e + r_0^{RP} - r_y \cdot \left(Y - \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi^e - \phi_\pi \cdot \left(\pi - \pi^T \right) \right] \right) \right] \\
\Leftrightarrow \quad (1 - IS_1 \cdot r_y) \cdot Y &= IS_0 - IS_1 \cdot \left(-\pi^e + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi^e - \phi_\pi \cdot \left(\pi - \pi^T \right) \right] \right) \\
\Leftrightarrow \quad (1 - IS_1 \cdot r_y) \cdot Y &= \frac{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}{\phi_y} \cdot \pi \\
&\quad + IS_0 - IS_1 \cdot \left[-\pi^e + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi^e + \phi_\pi \cdot \pi^T \right) \right]
\end{aligned}$$

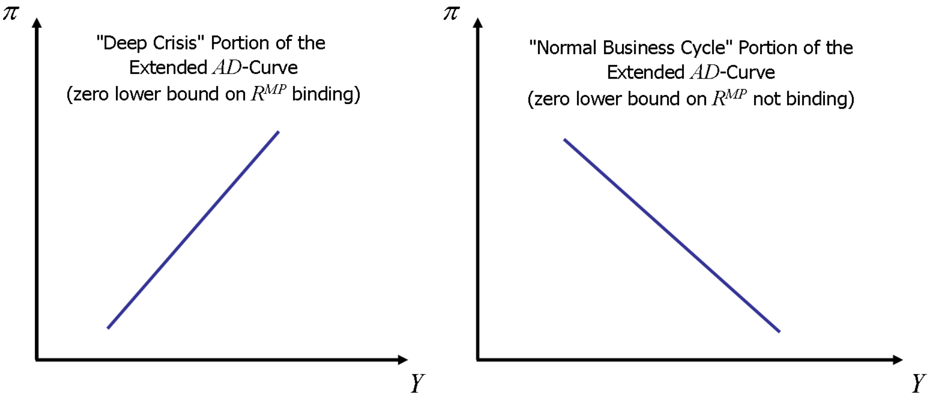
$$\pi = \left[\frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right] \cdot Y - \left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot \left(IS_0 - IS_1 \cdot \left[-\pi^e + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi^e + \phi_\pi \cdot \pi^T\right)\right]\right) \tag{30}$$

+ Note that the **"deep crisis" portion of the extended *AD*-curve** in Equation (30) exhibits (for empirically sensible parameter values, including $0 < IS_1 \cdot r_y < 1$) a **positive relationship between output, Y, and inflation, π**, and is thus **upward sloping**.
+ This reflects that as the rate of inflation decreases, $Y^{ZLB}|_\pi$ from Equation (25) increases (as the central bank for "normal business cycle" outcomes then levies lower monetary policy rates), implying that
    - the endogenous component of the risk premium becomes positive at higher levels of output than before, and
    - for any level of output that is below the level of $Y^{ZLB}|_\pi$ that prevailed prior to the decrease of the rate of inflation, the endogenous component of the risk premium is now higher than before, in turn crowding out more of aggregate demand.

Let us illustrate graphically also that the "deep crisis" portion of the extended *AD*-curve is upward sloping:

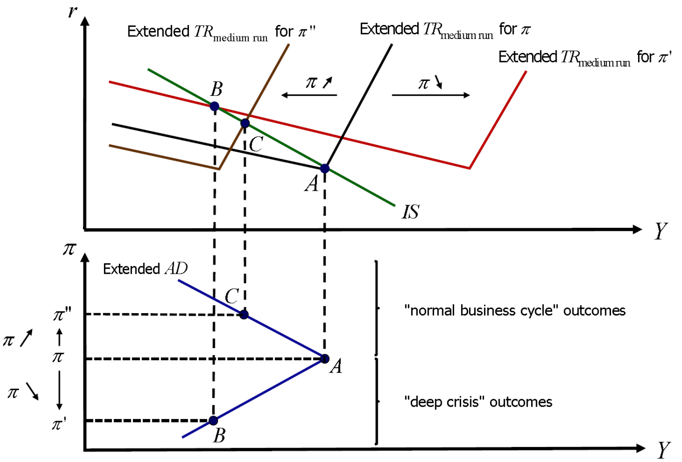

Extended $TR_{\text{medium run}}$ for π
Extended $TR_{\text{medium run}}$ for π'
r
IS
$\pi' < \pi$
B
A
$\pi \searrow$
Y
$Y^{ZLB}|_{\pi}$
$Y^{ZLB}|_{\pi'}$
π
"Deep Crisis" Portion of the Extended AD-Curve
(zero lower bound on R^{MP} binding)
π
$\pi \searrow$
π'
A
B
Y'
Y
Y

How to combine the ”deep crisis” and the ”normal business cycle” portions of the extended AD-curve graphs into a single extended AD-curve graph?



- We have already seen that the zero lower bound level of output depends on the rate of inflation. It is thus plausible to expect that in the extended *AD*-curve diagram, whether the *AD*-curve reflects a ”normal business cycle” or a ”deep crisis” outcome depends on the rate of inflation.
- When the rate of inflation is relatively low, from the Taylor Rule, the nominal monetary policy rate *ceteris paribus* is relatively low, rendering it more likely that we may reach the zero lower bound, $R^{MP} = 0$.
- When the rate of inflation is relatively high, from the Taylor Rule, the nominal monetary policy rate *ceteris paribus* is relatively high, rendering it less likely that we may

The following graph shows that this line of reasoning is correct:



Let us inspect the **extended *AD*-curve** in more detail:

+ At which rate of inflation does the extended *AD*-curve switch from being downward sloping ("normal business cycle" outcomes, $R^{MP} > 0$) to being upward sloping ("deep crisis" outcomes, $R^{MP} = 0$)?
+ The discontinuity in the extended *AD*-curve occurs at the highest rate of inflation at which the zero lower bound on R^{MP} is binding. From the extended monetary policy rule, Equation (23), this rate of inflation depends on the level of output. To indicate this dependence, we denote the zero lower bound rate of inflation as $\pi^{ZLB}|_Y$:

$$r_0^{MP} + \pi^e + \phi_Y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi^{ZLB}|_Y - \pi^T\right) = 0$$

$$\Leftrightarrow \quad \pi^{ZLB}|_Y = \pi^T - \frac{1}{\phi_\pi} \cdot \left[r_0^{MP} + \pi^e + \phi_Y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right)\right]. \tag{31}$$

+ To obtain the unconditional π^{ZLB}, we need to eliminate Y in Equation (31).
+ Since at the point of discontinuity in the extended *AD*-curve it not only holds that $R^{MP} = 0$, but it also holds that the endogenous component of the risk premium is equal to zero, from the medium-run extended *TR*-curve, Equations (27) and (28), we have that at the point of discontinuity in the extended *AD*-curve it holds that

$$r = -\pi^e + r_0^{RP}. \tag{32}$$

+ From the *IS*-curve relationship in Equation (12) we therefore have that at the point of discontinuity in the extended *AD*-curve it holds that

$$Y = Y^{ZLB} = IS_0 - IS_1 \cdot \left(-\pi^e + r_0^{RP}\right). \tag{33}$$

+ Substituting from Equation (33) back into the right-hand side of Equation (31), we have that at the point of discontinuity in the extended *AD*-curve it also holds that:

$$\pi^{ZLB} = \pi^T - \frac{1}{\phi_\pi} \cdot \left(r_0^{MP} + \pi^e + \frac{\phi_Y}{\bar{Y}} \cdot \left[IS_0 - IS_1 \cdot \left(-\pi^e + r_0^{RP}\right)\right] - \phi_Y\right). \tag{34}$$

- Note from Equations (33) and (34) that both Y^{ZLB} and π^{ZLB} are functions of the expected rate of inflation, π^e. Note furthermore from Equation (30) that the intercept of the "deep crisis" portion of the extended *AD*-curve also is a function of π^e.
- Therefore, as under adaptive inflation expectations we know from the *AS-AD* model that the medium-run adjustment paths following, say, a stimulus to aggregate demand can involve continued changes in π^e, such changes then lead to continued shifts of the extended *AD*-curve. This will be important to keep in mind when analyzing medium-run adjustment paths in the extended *AS-AD* model.

To understand what outcomes may constitute a "deep crisis" outcome, it will be useful to close our discussion of the extended *AD*-curve with a **numerical example**:

- Suppose $A = 9.78$, $C_y = 0.6$, $G_y = 0.15$, $TB_{im} = 0.05$, $TB_{ex} = 0.01$, $C_r = 4$, $I_r = 16$, $TB_\varepsilon = 1$, $\varepsilon_r = 6$, $\bar{Y} = 15$, $r_0^{MP} = 0.02$, $r_0^{RP} = 0.01$, $\phi_y = 0.5$, $\phi_\pi = 2$, and $\pi^T = 0.02$.
- Substituting these parameter values into Equations (33) and (34), using the definitions of IS_0 and IS_1 given in Equation (12), and under the further specification $\pi^e = \pi^T$, we obtain $Y^{ZLB} \approx 16.733$ and $\pi^{ZLB} \approx -0.029$.
- This numerical example thus illustrates that in the extended *AS-AD* model (unlike in the extended *IS-TR* model for the macroeconomy in the short run), a "deep crisis" outcome does not have to be a crisis outcome in terms of the level of output. Rather (albeit hypothetically), the central bank may set $R^{MP} = 0$ even though actual output is above the long-run level of output, and does so because actual inflation is well below the target rate of inflation (and is possibly negative).

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- A "deep crisis" outcome still can be a crisis outcome in terms of the level of output: Under the same parameter values as above except that we had that $\pi^e = \pi^T - 0.06$, we would obtain $Y^{ZLB} \approx 14.133$ and $\pi^{ZLB} \approx 0.044$.
- This latter numerical example thus illustrates that (hypothetically) the central bank may set $R^{MP} = 0$ even though actual inflation is above the target rate of inflation, and does so because actual output is well below the long-run level of output.
- As we will see when analyzing interesting adjustment paths in the extended *AS-AD* model, the key magnitude to consider then is $Y^{ZLB}\big|_{\pi}$, and the conditional levels

Workbook for the Extended AD-Curve

The workbook *Extended-AD-Curve.xlsm* allows to study all the determinants of the position and slope of the extended AD-curve.

Extended AD-Curve

Specification

Extended AD-Curve Parameters

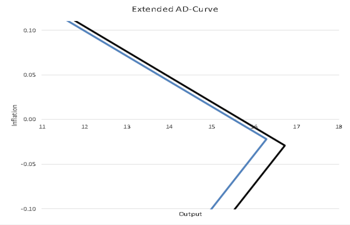
Extended TR-Curve Parameters	Initial	New
r_0^{MP}	0.020	0.020
r_0^{RP}	0.010	0.020
r_y	0.005	0.005
ϕ_y	0.500	0.500
ϕ_π	2.000	2.000
$\bar{Y}$	15.000	15.000
π^T	0.020	0.020
π^e	0.020	0.020

IS-Curve Parameters	Initial	New
C_r	4.000	4.000
I_r	16.000	16.000
TB_ε	1.000	1.000
ε_r	6.000	6.000
C_y	0.600	0.600
TB_{im}	0.050	0.050
G_y	0.150	0.150
A	9.780	9.780
C_0	1.500	1.500
I_0	3.000	3.000
G_0	2.800	2.800
TB_0	0.000	0.000
ε_0	1.000	1.000
T	3.000	3.000
TB_{ex}	0.010	0.010
Y^*	100.000	100.000
T^*	30.000	30.000
r^*	0.030	0.030

Quantitative Results

	Initial	New
π^{ZLB}	-0.029	-0.022

Graph



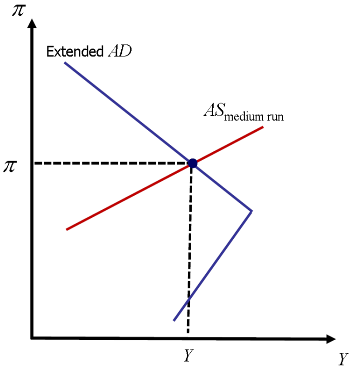
Medium-run macroeconomic outcomes in the **extended *AS-AD* model**, that graphically occur at the intersection of the extended *AD*-curve with the medium-run *AS*-curve, may materialize either at

- a combination of inflation and output for which we are in a "normal business cycle" outcome,

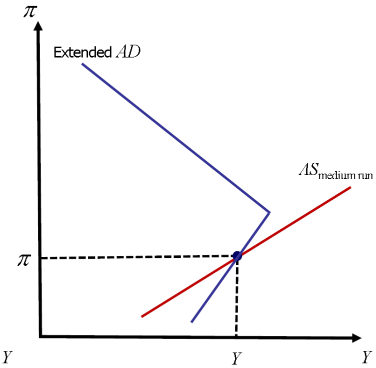
or

- a combination of inflation and output for which we are in a "deep crisis" outcome.

The following graph illustrates this:



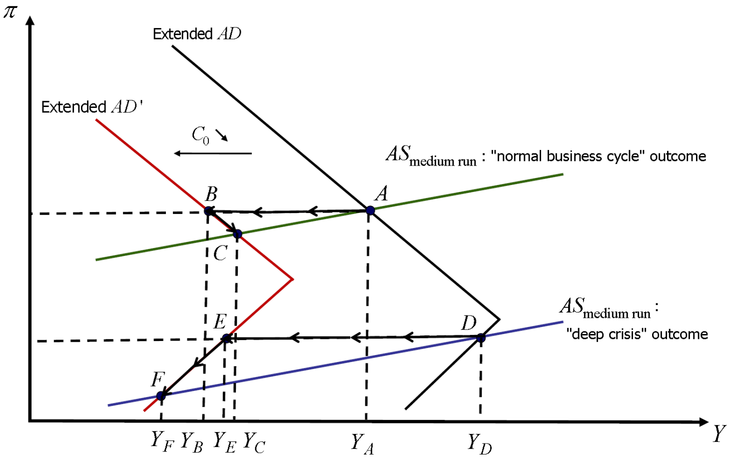
"normal business cycle" outcome (along the downward-sloping portion of the extended AD-curve)



"deep crisis" outcome (along the upward-sloping portion of the extended AD-curve)

Can the extended *AS-AD* model explain severe *and* prolonged recessions, such as the one that occurred in the wake of the global financial and European sovereign debt crises, and if so, what are the extended *AS-AD* model's novel propagation channels relative to those of the standard *AS-AD* model?

To address these questions, let us examine first how the magnitudes of the losses of output in the extended *AS-AD* model that occur after a drop in the autonomous component of aggregate demand (say, a collapse in housing prices, implying $C_0 \searrow$) compare across the "normal business cycle" and "deep crisis" outcomes. To keep this analysis brief, let us suppose for purposes of this comparison that inflation expectations are fixed:



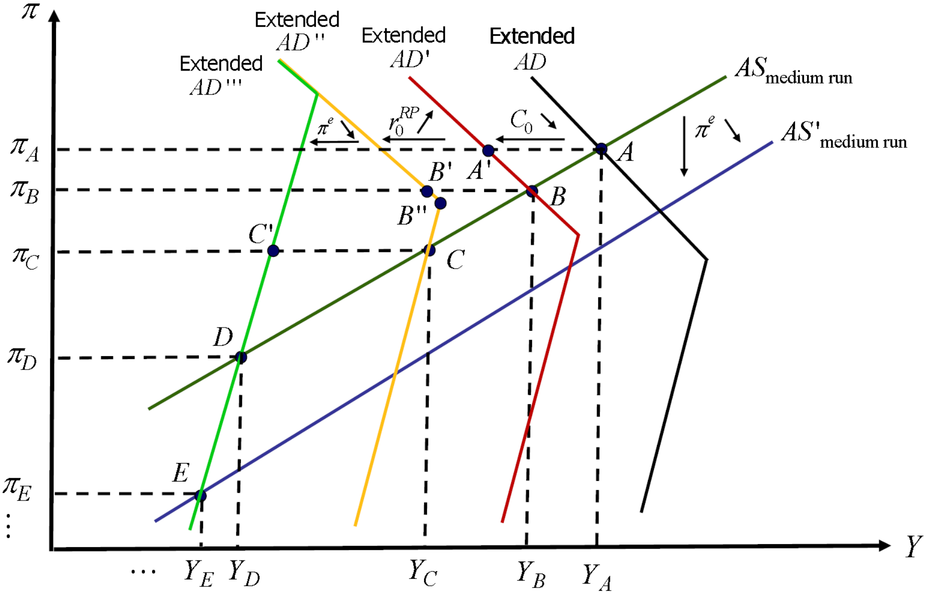
+ In regards to the shift of the extended *AD*-curve, following the decrease in C_0 (that in turn leads to a decrease of IS_0), for a given rate of inflation there is a lower level of aggregate demand and thus output, and the extended *AD*-curve generally shifts to the left. As for the point of discontinuity in the extended *AD*-curve, from (33) we observe that Y^{ZLB} must decrease, and from (34) we observe that π^{ZLB} must increase. Also note from Equations (14) and (30) that the slopes of both portions of the extended *AD*-curve do not depend on C_0 and thus remain unchanged.
+ Note that when we are not in the "deep crisis" case, the decrease in C_0 through the lower level of aggregate demand for a given rate of inflation leads to a decrease in output (including through Keynesian multiplier effects, that are only partially offset by the central bank lowering the monetary policy rate in response to the decrease in output) (movement from Point *A* to Point *B*).
+ As output falls, the wage mark-ups and subsequently inflation fall as well. The central bank will thus lower the monetary policy rate in response also to the decrease in the rate of inflation, leading to rises of consumption, investment and the trade balance, that partially offset the initial decline in output (movement from Point *B* to Point *C*).
+ These various monetary policy rate adjustments are absent in the "deep crisis" case (due to the zero lower bound constraint being binding), and this is the first reason why the loss of output following the decrease in C_0 is more severe in the "deep crisis" case than in the "normal business cycle" case.
+ The initial fall of output in the "deep crisis" case (movement from Point *D* to Point *E*) reflects the decrease in C_0, augmented by Keynesian multiplier effects and a rise of the endogenous component of the risk premium, leading (despite monetary policy operating at the zero lower bound) to an increase of the interest rate at which households and firms can borrow, and thus crowding out of consumption, investment and the trade balance.

+ As output falls, again the wage mark-ups and subsequently inflation fall as well. $Y^{ZLB}\mid_{\pi}$ increases, implying that the endogenous component of the risk premium rises yet further (it becomes positive at higher levels of output than before), and for any level of output that is below the level of $Y^{ZLB}\mid_{\pi}$ that prevailed prior to the decrease of the rate of inflation, the endogenous component of the risk premium is now higher than before, crowding out yet more of aggregate demand (movement from Point *E* to Point *F*).
+ The various increases in the endogenous component of the risk premium are a propagation mechanism only present in the "deep crisis" case, and are the second reason why the loss of output following the decrease in C_0 is more severe in the "deep crisis" case than in the "normal business cycle" case.

To see that the extended *AS*-*AD* model can explain recessions that are not just deeper, but also more prolonged than those in the standard *AS*-*AD* model, let us examine within the extended *AS*-*AD* model the medium-run macroeconomic effects of the two shocks that seemed so important for the global financial crisis:

- a collapse in housing prices, implying $C_0 \searrow$, coupled with
- an increase in the exogenous component of the risk premium, $r_0^{RP} \nearrow$.

To provide complete reasoning, we now take into account again that inflation expectations are adaptive.



π
Extended AD''
Extended AD'
Extended AD
Extended AD'''
$AS_{\text{medium run}}$
$AS'_{\text{medium run}}$
π^e
r_0^{RP}
C_0
π^e
π_A
π_B
π_C
π_D
π_E
A
A'
B
B'
B''
C
C'
D
E
Y
$\cdots$ Y_E Y_D Y_C Y_B Y_A

- As noted before, following the decrease of C_0 (that in turn leads to a decrease of IS_0), the extended AD-curve shifts to the left and upwards, as the point of discontinuity of the extended AD-curve shifts to the left and upwards, and the slopes of both portions of the extended AD-curve remain unchanged.
- Following the shift of the extended AD-curve, for a given rate of inflation there is now a lower level of aggregate demand and thus output (initial drop in aggregate demand plus Keynesian multiplier effects), partially offset by a crowding in of consumption, investment and the trade balance, as the central bank in response to the decrease in output lowers the monetary policy rate (from Point A to Point A').
- As output falls, wage mark-ups and subsequently inflation fall as well. The central bank lowers the monetary policy rate further, inducing further crowding in (from Point A' to Point B).
- Following the increase of r_0^{RP}, the extended AD-curve shifts to the left and upwards: from (33) we observe that Y^{ZLB} decreases, and from (34) we observe that π^{ZLB} increases. The point of discontinuity of the extended AD-curve thus again shifts to the left and upwards.
- As r_0^{RP} increases, so does the real interest rate, causing, for a given rate of inflation, crowding out of consumption, investment and the trade balance (from Point B to Point B').
- As output falls further, wage mark-ups and subsequently inflation fall further as well. The central bank lowers the monetary policy rate further, causing partial crowding in of consumption, investment and the trade balance, but, as graphed, hits the zero lower bound in the process (from Point B' to Point B'').
- At Point B'', the economy has moved into a "deep crisis" outcome. As the rate of inflation keeps falling in the adjustment towards the medium-run AS-curve, the endogenous component of the risk premium comes into effect and then increases, reflecting continued increases in $Y^{ZLB}\big|_{\pi}$, as per Equation (25). (As the zero lower bound for the monetary policy rate becomes effective at increasingly higher levels of output, the endogenous component of the risk premium keeps increasing.)

+ Under adaptive inflation expectations overall medium-run adjustment does not stop at Point C, though (it would stop there if inflation expectations were static).
+ **Movement from Point C to Point D:**
    - Under adaptive inflation expectations, given the decrease in the actual rate of inflation from Point A to Point C, inflation expectations will be revised downward in the second period of adjustment. While this does not affect the downward-sloping portion of the extended AD-curve (see (14)), the intercept of the upward-sloping portion of the extended AD-curve increases (see (30)). Also, from (33) we observe that Y^{ZLB} decreases, and from (34) we observe that π^{ZLB} increases. The point of discontinuity of the extended AD-curve thus once more shifts to the left and upwards.
    - As π^e decreases, the "deep crisis" real interest rate increases further, recalling from the medium-run extended TR-curve, Equation (27), that

$$r = -\pi^e + r_0^{RP} - r_y \cdot \left(Y - Y^{ZLB}\Big|_{\pi} \right).$$

    This further increase of the real interest rate for a given actual rate of inflation causes further crowding out of consumption, investment and the trade balance (from Point C to Point C').
+ As the actual rate of inflation keeps falling in the adjustment towards the medium-run AS-curve, the endogenous component of the risk premium increases further, reflecting continued increases in $Y^{ZLB}\Big|_{\pi}$, as per Equation (25). (As the zero lower bound for the monetary policy rate becomes effective at increasingly higher levels of output, the endogenous component of the risk premium increases further.)
+ There is thus yet further crowding out of consumption, investment and the trade balance in the movement from Point C' to Point D.

+ **Movement from Point *D* to Point *E*:**
    - Not to be neglected, the decrease of inflation expectations in the second period of adjustment also causes a downward shift of the medium-run *AS*-curve (for a given level of output, actual inflation decreases, as wage growth and thus the growth rate of production costs fall).
    - As the rate of inflation keeps falling in the adjustment towards the new medium-run *AS*-curve, the endogenous component of the risk premium increases yet further, reflecting further continued increases in $Y^{ZLB}\mid_{\pi}$, as per Equation (25). (As the zero lower bound for the monetary policy rate becomes effective at yet increasingly higher levels of output, the endogenous component of the risk premium increases yet further.)
    - There is thus yet further crowding out of consumption, investment and the trade balance in the movement from Point *D* to Point *E*. Point *E* is the medium-run outcome in both the real and the financial sector in the second period of adjustment.
+ Full medium-run adjustment does not end at Point *E*, as under adaptive inflation expectations the spiral of decreases in the actual rate of inflation causing in the following period further decreases in inflation expectations (implying yet higher real interest rates), that in turn imply a yet lower actual rate of inflation (with the latter leading to yet further rises of the endogenous component of the risk premium and thus again of the real interest rate), keeps operating.
+ In comparison to the standard *AS-AD* model, then, the extended *AS-AD* model can not only explain deeper recessions (the various increases in the endogenous component of the risk premium are a propagation mechanism that is only present for "deep crisis" outcomes), but can also explain more prolonged recessions (reflecting that for "deep crisis" outcomes the decreases of inflation expectations that occur following losses of output lead to further losses of output, and thus mutually re-inforcing spirals of decreases of output, actual inflation and expected inflation can arise).

+ As was the case in the short-run analysis involving the extended *IS-TR* model, a complication in the quantitative analysis of the extended *AS-AD* model is that (in general) we do not know prior to our calculations whether the medium-run outcome is a "normal business cycle" outcome or a "deep crisis" outcome.
+ Suppose initially, though, that we knew that the outcome was a **"normal business cycle" outcome**, with $R_t^{MP} > 0$. We already derived the level of output holding in this case when considering the standard *AS-AD* model, see in particular Equation (20):

$$\begin{aligned} Y_t &= \frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \pi_{t-1} + \omega - s_t \\ &= \frac{\omega}{\bar{Y}} + \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \end{aligned}$$

+ The rate of inflation in the "normal business cycle" medium-run outcome can in turn be obtained from Equation (21),

$$\pi_t = \pi_{t-1} - \omega + s_t + \frac{\omega}{\bar{Y}} \cdot Y_t,$$

with Y_t given by Equation (20).
+ Using these levels of output and the rate of inflation, the real interest rate in the "normal business cycle" medium-run outcome from Equation (22) is equal to

$$r_t = r_0^{MP} + \phi_y \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi_t - \pi^T\right) + r_0^{RP}.$$

+ The nominal monetary policy rate corresponding to this real interest rate from the medium-run extended *TR*-curve in Equation (27), amended to include time subscripts and accounting for adaptive expectations, is given by

$$R_t^{MP} = r_0^{MP} + \pi_{t-1} + \phi_y \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}} \right) + \phi_\pi \cdot \left(\pi_t - \pi^T \right). \qquad (35)$$

+ Given output, inflation and the real interest rate, the "normal business cycle" medium-run macroeconomic outcomes of the other key macroeconomic aggregates, including consumption, investment, government expenditure, the trade balance and the exchange rate, may be retrieved from the *IS*-portion of the model.
+ Suppose next that we knew that the outcome was a **"deep crisis" outcome**, with the zero lower bound constraint on the monetary policy rate binding, $R_t^{MP} = 0$.
+ To derive the level of output holding in this case, we set the rate of inflation as implied by the medium-run *AS*-curve, Equation (17), equal to the rate of inflation implied by the "deep crisis" portion of the extended *AD*-curve, Equation (30), amended to include time subscripts and to incorporate adaptive expectations,

$$\underbrace{\pi_t = \pi_{t-1} + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}} \right) + s_t}_{\text{medium-run } AS\text{-curve (17)}} = \underbrace{\left[\frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi} \right] \cdot Y_t - \left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi} \right) \cdot (IS_0 - IS_1)}_{\text{from "deep crisis" portion of extended } AD\text{-curve (30)}}$$

$$\cdot \left[-\pi_{t-1} + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi_{t-1} + \phi_\pi \cdot \pi^T \right) \right]$$

- Note that (36) is one equation in one unknown, namely Y_t. We can thus use Equation (36) to solve for Y_t:

$$\left[\frac{\omega}{\bar{Y}} - \frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right] \cdot Y_t$$

$$= -\left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot (IS_0 - IS_1) \cdot \left[-\pi_{t-1} + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi_{t-1} + \phi_\pi \cdot \pi^T\right)\right] - \pi_{t-1} + \omega - s_t$$

$$\Leftrightarrow \quad Y_t = \frac{\left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot (IS_0 - IS_1) \cdot \left[-\pi_{t-1} + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi_{t-1} + \phi_\pi \cdot \pi^T\right)\right] + \pi_{t-1} - \omega + s_t}{\frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi} - \frac{\omega}{\bar{Y}}} \tag{37}$$

- **Equation (37)** gives us the level of **output** in the ”deep crisis” medium-run macroeconomic outcome.
- Having obtained the level of output, we can calculate the rate of **inflation** in the ”deep crisis” medium-run macroeconomic outcome from the medium-run *AS*-curve, Equation (17), to obtain, as in Equation (21),

$$\pi_t = \pi_{t-1} - \omega + s_t + \frac{\omega}{\bar{Y}} \cdot Y_t,$$

with Y_t given by Equation (37).

+ The **real interest rate** in the "deep crisis" medium-run macroeconomic outcome may be obtained from the medium-run extended *TR*-curve, Equations (27) and (28), amended to include time subscripts and accounting for adaptive expectations, as

$$r_t = -\pi_{t-1} + r_0^{RP} - r_y \cdot \left(Y_t - Y_t^{ZLB} \Big|_{\pi} \right) \tag{38}$$

with Y_t given by Equation (37), and with $Y_t^{ZLB}\Big|_{\pi}$ given by Equation (25), also amended to include time subscripts and accounting for adaptive expectations,

$$Y_t^{ZLB}\Big|_{\pi} = \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi_{t-1} - \phi_\pi \cdot (\pi_t - \pi^T) \right], \tag{39}$$

where π_t is given by Equation (21).

+ Given output, inflation and the real interest rate, the "deep crisis" medium-run macroeconomic outcomes of the other key macroeconomic aggregates, including consumption, investment, government expenditure, the trade balance and the exchange rate, may be retrieved from the *IS*-portion of the model.

+ Suppose that we start out in period $t=0$ in the long-run macroeconomic outcome where $Y_0=\bar{Y}$, $r_0=r_0^{MP}+r_0^{RP}$, and $\pi_0=\pi^T$.
+ Also suppose $A=9.78$, $C_y=0.6$, $G_y=0.15$, $TB_{im}=0.05$, $TB_{ex}=0.01$, $C_r=4$, $I_r=16$, $TB_\varepsilon=1$, $\varepsilon_r=6$, $\bar{Y}=15$, $r_0^{MP}=0.02$, $r_0^{RP}=0.01$, $\phi_y=0.5$, $\phi_\pi=2$, $r_y=0.005$, $\pi^T=0.02$, $\omega=0.05$ and $s=0$.
+ Note that (in general) we will not know prior to our calculations whether the medium-run macroeconomic outcomes are "normal business cycle" outcomes or "deep crisis" outcomes.
+ One approach to resolving this complication is as follows:
+ ***Step 1:*** Conjecture that in period t we are in a "normal business cycle" outcome.
  - (We make this conjecture as in the medium-run analysis the algebra to solve the model is a bit less involved for the "normal business cycle" outcome than for the "deep crisis" outcome).
  - Based on this conjecture, calculate the implied level of output from Equation (20):

$$Y_t^{Conjecture}=\frac{\left(\phi_y+\phi_\pi\cdot\pi^T-r_0^{MP}-r_0^{RP}\right)\cdot IS_1+IS_0}{\phi_\pi\cdot IS_1}-\pi_{t-1}+\omega-S_t\Big/\frac{\omega}{\bar{Y}}+\frac{\bar{Y}+\phi_y\cdot IS_1}{\phi_\pi\cdot IS_1\cdot\bar{Y}}$$

+ ***Step 2:*** From the medium-run extended *TR*-curve in Equation (27), amended to include time subscripts and accounting for adaptive expectations,

$$r_t\approx\max\left\{-\pi_{t-1},\ r_0^{MP}+\phi_y\cdot\left(\frac{Y_t-\bar{Y}}{\bar{Y}}\right)+\phi_\pi\cdot\left(\pi_t-\pi^T\right)+r_0^{RP}-r_y\cdot\left(Y_t-Y_t^{ZLB}\Big|_{\pi_t}\right)\cdot I\left(Y_t\leq Y_t^{ZLB}\Big|_{\pi_t}\right)\right\},$$

  - it is clear that whether in period t we are in a "normal business cycle" or a "deep crisis" outcome for a given level of output depends on the highest level of output at which the zero lower bound on the monetary policy rate is binding, that is, $Y_t^{ZLB}\big|_{\pi_t}$.

- Using $Y_t^{Conjecture}$ calculated in Step 1, we thus calculate $Y_t^{ZLB}\big|_{\pi_t^{Conjecture}}$ from Equation (39) as

$$Y_t^{ZLB}\Big|_{\pi_t^{Conjecture}} = \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \pi_{t-1} - \phi_\pi \cdot \left(\pi_t^{Conjecture} - \pi^T\right)\right],$$

- where $\pi_t^{Conjecture}$ from Equation (21) is given by

$$\pi_t^{Conjecture} = \pi_{t-1} - \omega + s_t + \frac{\omega}{\bar{Y}} \cdot Y_t^{Conjecture}.$$

+ **Step 3:** If $Y_t^{Conjecture}$ (calculated in Step 1) was larger than $Y_t^{ZLB}\big|_{\pi_t^{Conjecture}}$ (calculated in Step 2),

$$Y_t^{Conjecture} > Y_t^{ZLB}\Big|_{\pi_t^{Conjecture}},$$

  we know that in period t we are indeed in a "normal business cycle" outcome, for which it holds that R_t^{MP} is not constrained at the zero lower bound, that is, from the Taylor rule in Equation (35) we have that

$$R_t^{MP} = r_0^{MP} + \pi_{t-1} + \phi_y \cdot \left(\frac{Y_t^{Conjecture} - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi_t^{Conjecture} - \pi^T\right) > 0.$$

+ We can then calculate the outcomes for all variables based on the relations that hold for the "normal business cycle" outcome, including the real interest rate from Equation (22), using that $Y_t = Y_t^{Conjecture}$ and that $\pi_t = \pi_t^{Conjecture}$ (note that $\pi_t^{Conjecture}$ was calculated in Step 2).

+ **Step 4:** If, however, the conjectured level of output calculated in Step 1 was less than or equal to Y_t^{ZLB},

$$Y_t^{Conjecture} \leq Y_t^{ZLB}\Big|_{\pi_t^{Conjecture}},$$

then a contradiction has arisen, and the period t outcome must be a “deep crisis” outcome, so that we need to calculate all outcomes based on the relations that hold when the zero lower bound on the monetary policy rate is binding:

- In particular, we can need to calculate medium-run output from Equation (37),

$$Y_t = \frac{\left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot (IS_0 - IS_1) \cdot \left[-\pi_{t-1} + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_0^{MP} - \pi_{t-1} + \phi_\pi \cdot \pi^T\right)\right] + \pi_{t-1} - \omega + s_t}{\frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi} - \frac{\omega}{\bar{Y}}}$$

- Using this level of output, we can then calculate the rate of inflation from Equation (21) and the real interest rate from Equation (38).

+ Presume at first that the period $t=2$ medium-run macroeconomic outcome is a “normal business cycle” outcome. From Equation (20), noting that following the two shocks to the consumption function and the risk premium we have that $r_0^{RP} = 0.05$ and

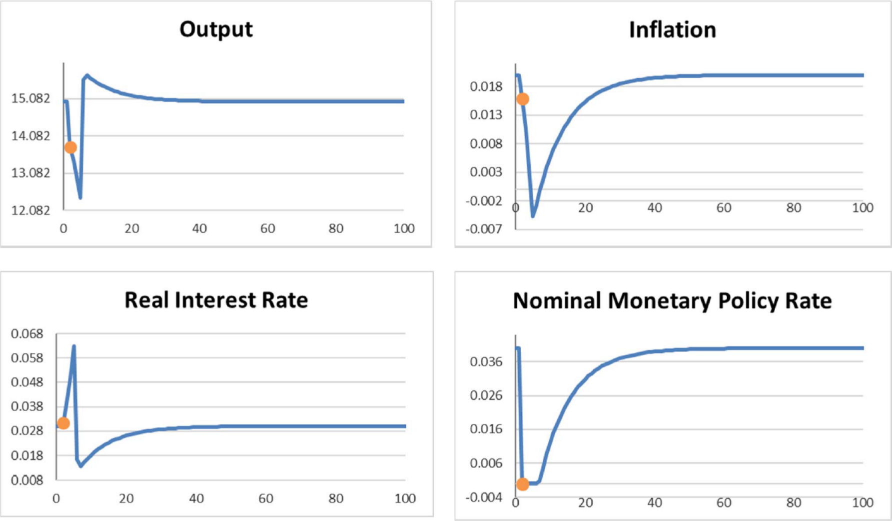
$$IS_0 = (A + \Delta C_0) \big/ [1 - (C_y - TB_{im} - G_y)] \approx 15.133,$$

we obtain $Y_2^{Conjecture} \approx 13.939$. From Equation (21) (and using the exact level of $Y_2^{Conjecture}$), we obtain that $\pi_2^{Conjecture} \approx 0.016$. Using the (exact) levels of $Y_2^{Conjecture}$ and of $\pi_2^{Conjecture}$ in Equation (39), we obtain that

$$13.939 \approx Y_2^{Conjecture} < Y_2^{ZLB}\Big|_{\pi_2^{Conjecture}} \approx 14.012.$$

We thus cannot be in a “normal business cycle” outcome in period $t=2$, and must rather be in a “deep crisis” outcome.

- The fact that the period $t=2$ outcome must be a “deep crisis” outcome may also be confirmed from the Taylor rule: Using the (exact) levels of $Y_2^{Conjecture}$ and of $\pi_2^{Conjecture}$ in Equation (35), we obtain that $R_2^{MP} \approx -0.002 < 0$.
- Calculating the level of output in period $t=2$ from Equation (37), we obtain $Y_2 \approx 13.775$. From Equation (21) (and using the exact level of Y_2), it then follows that $\pi_2 \approx 0.016$. From Equation (35), the Taylor rule for these values of Y_2 and π_2 would imply that $R_2^{MP} \approx -0.009$, once again confirming that the zero lower bound restriction is binding in period $t=2$.
- The following graphs show the full medium-run adjustment paths implied by the extended *AS-AD* model, based on automated calculations within the workbook *Extended-AS-AD-Model-FinancialCrises.xlsm*, and presuming that from period $t = 6$ onwards, the decrease of the intercept of the consumption function and the increase of the exogenous component of the risk premium are reversed: relative to periods $t = 2$ to $t = 5$, we have that $\Delta C_0 = 0.7$ and $\Delta r_0^{RP} = -0.04$, so that from period six onwards all parameter values are again as in period one.
- Using the workbook *Extended-AS-AD-Model-FinancialCrises.xlsm*, these dynamics can also be traced within the extended *AS-AD* model diagram.


Output
15.082
14.082
13.082
12.082
0
20
40
60
80
100
Inflation
0.018
0.013
0.008
0.003
-0.002
-0.007
0
20
40
60
80
100
Real Interest Rate
0.068
0.058
0.048
0.038
0.028
0.018
0.008
0
20
40
60
80
100
Nominal Monetary Policy Rate
0.036
0.026
0.016
0.006
-0.004
0
20
40
60
80
100

Workbook for the Extended *AS-AD* Model

The workbook *Extended-AS-AD-Model-FinancialCrises.xlsm* allows to study how changes in the various parameters and exogenous variables of the *Extended-AS-AD*-model …

Extended AS-AD Model (Financial Crises): Parameter Specification

	TR-Curve Parameters							IS-Curve Parameters																		AS-Curve Parameters	
Period	r_0^{MP}	r_0^{RP}	r_y	ϕ_y	ϕ_π	$\bar{Y}$	π^T	c_r	i_r	TB_ε	ε_r	c_y	TB_{im}	G_y	A	C_0	I_0	G_0	TB_0	ε_0	T	TB_{ex}	Y^*	T^*	r^*	ω	s
0	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
2	0.020	0.050	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.080	0.800	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
3	0.020	0.050	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.080	0.800	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
4	0.020	0.050	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.080	0.800	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
5	0.020	0.050	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.080	0.800	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
6	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
7	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
8	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
9	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
10	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
11	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
12	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
13	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
14	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
15	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
16	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
17	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
18	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
19	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
20	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
21	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
22	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
23	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
24	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
25	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
26	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
27	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
28	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
29	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
30	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
31	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
32	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
33	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
34	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
35	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
36	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
37	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
38	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
39	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
40	0.020	0.010	0.005	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000

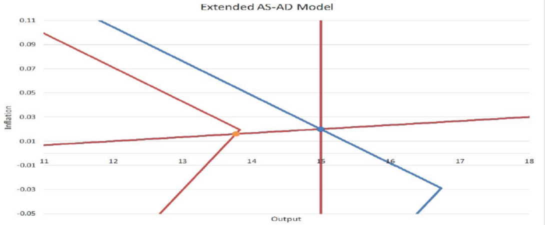
... affect output, inflation and the other endogenous variables in the medium-run macroeconomic outcome, ...

Extended AS-AD Model (Financial Crises): Quantitative Results and Extended AS-AD Diagram

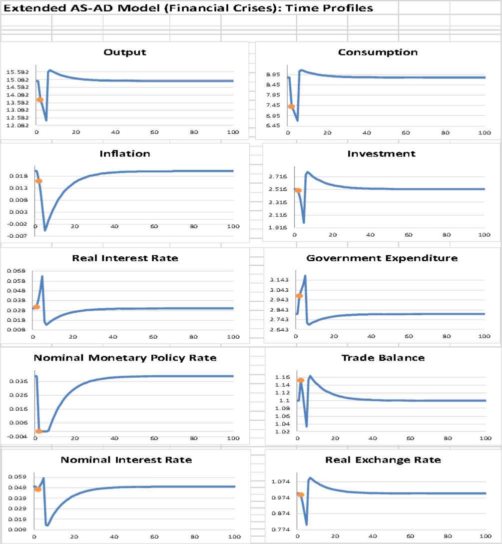
Medium-Run Macroeconomic Outcomes

	Period 0	Period 1	Period 2	Period 1000
Inflation	0.020	0.020	0.016	0.020
Output	15.000	15.000	13.775	15.000
Real Interest Rate	0.030	0.030	0.031	0.030
Consumption	8.820	8.820	7.390	8.820
Investment	2.520	2.520	2.498	2.520
Government Expenditure	2.800	2.800	2.984	2.800
Trade Balance	1.100	1.100	1.153	1.100
Real Exchange Rate	1.000	1.000	0.992	1.000
Nominal Monetary Policy Rate	0.040	0.040	0.000	0.040
Nominal Interest Rate	0.050	0.050	0.047	0.050

	Current	
Period	2	Plus / Minus



… and also graphs the transition dynamics (from the initialization onwards, and depicting the full medium-run adjustment):



+ For the macroeconomy in the short run, we analyzed how the outbreak of the COVID-19 pandemic affected aggregate economic activity through a combination of aggregate demand and supply shocks, specifically
  - (i) an increase in the fraction of consumption expenditure not occurring due to lockdowns and/or social distancing,
  - (ii) an increase in the fraction of exports not occurring due to lockdowns and/or social distancing abroad, and
  - (iii) a decrease of productivity.
+ Let us now turn to studying the effects of these shocks on medium-run macroeconomic outcomes. We consider the following extended *AS-AD* model that maintains all features of the model that we used to study the medium-run macroeconomic impact of financial crises, and augment it with
  - (i) an aggregate consumption function allowing for some fraction of consumption expenditure not occurring due to lockdowns and/or social distancing,

$$C = (1 - \iota) \cdot \left[C_0 + C_y \cdot (Y - T) - C_r \cdot r\right],$$

    see Equation (107) of the chapter on "The Macroeconomy in the Short Run";
  - (ii) a trade balance equation allowing for corresponding effects of foreign lockdown restrictions and/or foreign social distancing on the exports,

$$TB = TB_0 + TB_{ex} \cdot (1 - i^*) \cdot (Y^* - T^*) - TB_{im} \cdot (1 - \iota) \cdot (Y - T) + TB_{\varepsilon} \cdot \left[\varepsilon_0 - \varepsilon_r \cdot (r - r^*)\right],$$

    see Equation (108) of the chapter on "The Macroeconomy in the Short Run".

- For the macroeconomy in the short run, we analyzed how the outbreak of the COVID-19 pandemic affected aggregate economic activity through a combination of aggregate demand and supply shocks, specifically (i) an increase in the fraction of consumption expenditure not occurring due to lockdowns and/or social distancing, (ii) an increase in the fraction of exports not occurring due to lockdowns and/or social distancing abroad, and (iii) a decrease of productivity.
- Let us now turn to studying the effects of these shocks on medium-run macroeconomic outcomes. We consider the following extended *AS-AD* model that maintains all features of the model that we used to study the medium-run macroeconomic impact of financial crises, and augment it with
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$$TB = TB_0 + TB_{ex} \cdot (1 - i^*) \cdot (Y^* - T^*) - TB_{im} \cdot (1 - \iota) \cdot (Y - T) + TB_\varepsilon \cdot \left[\varepsilon_0 - \varepsilon_r \cdot (r - r^*)\right],$$

  see Equation (108) of the chapter on "The Macroeconomy in the Short Run"; and
- (iii) a medium-run production function,

$$Y = f_0 + f_1 \cdot L + z,$$

  mirroring Equation (109) of the chapter on "The Macroeconomy in the Short Run", coupled with Equation (112) of the latter chapter,

$$r_n^{MP} = r_0^{MP} + \phi_z \cdot (z - \bar{z}), \tag{40}$$

- linking productivity shocks (z) to the neutral real monetary policy rate: the higher the level of firms' productivity, ceteris paribus the higher their demand for loans to fund purchases of physical capital, and the higher the rate of return on savings. The neutral monetary policy rate responds to such changes in the rate of return on savings along the long-run trend path of the macroeconomy.
- Combining the *IS*-curve in Equations (114) and (115) of the chapter on "The Macroeconomy in the Short Run",

$$Y = \frac{A^P}{1-[(1-t)\cdot(C_y - TB_{im}) - G_y]} - \frac{(1-t)\cdot C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1-[(1-t)\cdot(C_y - TB_{im}) - G_y]} \cdot r, \tag{41}$$

where

$$\begin{aligned} A^P = {} & (1-t)\cdot C_0 + I_0 + G_0 + TB_0 + TB_\varepsilon \cdot \varepsilon_0 - (1-t)\cdot(C_y - TB_{im})\cdot T \\ & + G_y \cdot \bar{Y} + TB_{ex}\cdot(1-i^*)\cdot(Y^* - T^*) + TB_\varepsilon \cdot \varepsilon_r \cdot r^*, \end{aligned} \tag{42}$$

- with the medium run extended *TR*-curve implied by Equation (27) and the specification of the neutral real monetary policy rate in Equation (40),

$$r = \max\left\{-\pi^e, \underbrace{r_0^{MP} + \phi_z\cdot(z-\bar{z})}_{r_n^{MP}} + \phi_y\cdot\left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi\cdot\left(\pi-\pi^T\right)\right\} + r_0^{RP} - r_y\cdot\left(Y - Y_t^{ZLB}\Big|_\pi\right)\cdot I\left(Y \le Y_t^{ZLB}\Big|\right. \tag{43}$$

+ we obtain the two portions of the extended *AD*-curve:
- $R^{MP} > 0$: "normal business cycle" outcomes (intermediate algebraic steps are as for the derivation of (14)):

$$\pi = \frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_n^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \cdot Y, \tag{44}$$

- $R^{MP} = 0$: "deep crisis" outcomes (intermediate algebraic steps are as for the derivation of (30)):

$$\begin{aligned}\pi = &\left(\frac{\phi_y \cdot (1 - IS_1 \cdot r_y)}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot Y \\ &- \left(\frac{\phi_y}{IS_1 \cdot r_y \cdot \bar{Y} \cdot \phi_\pi}\right) \cdot (IS_0 - IS_1) \cdot \left[-\pi^e + r_0^{RP} + \frac{r_y \cdot \bar{Y}}{\phi_y} \cdot \left(\phi_y - r_n^{MP} - \pi^e + \phi_\pi \cdot \pi^T\right)\right],\end{aligned} \tag{45}$$

- where now (from Equation (41))

$$IS_0 = \frac{A^P}{1 - [(1-t) \cdot (C_y - TB_{im}) - G_y]} \quad \text{and} \quad IS_1 = \frac{(1-t) \cdot C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - [(1-t) \cdot (C_y - TB_{im}) - G_y]}. \tag{46}$$

For what follows, it will also be useful to recall the following earlier results for the extended *AS-AD* model, updated to take account of the specification of the neutral real monetary policy rate in Equation (40):

+ The highest level of output at which the zero lower bound on R^{MP} is binding, conditional on the rate of inflation, may be derived in the same type of steps as (25), and is given by

$$Y_t^{ZLB}\Big|_\pi = \frac{\bar{Y}}{\phi_y} \cdot \left[\phi_y - r_0^{MP} - \phi_z \cdot (z - \bar{z}) - \pi^e - \phi_\pi \cdot \left(\pi - \pi^T\right)\right]. \tag{47}$$

+ Using the same type of steps as for the derivation of (33) and (34), the point of discontinuity of the extended *AD*-curve is given by

$$Y^{ZLB} = IS_0 - IS_1 \cdot \left(-\pi^e + r_0^{RP}\right), \tag{48}$$

and

$$\pi^{ZLB} = \pi^T - \frac{1}{\phi_\pi} \cdot \left(r_0^{MP} + \phi_z \cdot (z - \bar{z}) + \pi^e + \frac{\phi_y}{\bar{Y}} \cdot \left[IS_0 - IS_1 \cdot \left(-\pi^e + r_0^{RP}\right)\right] - \phi_y\right). \tag{49}$$

As in all of our prior analyses of the macroeconomy in the medium run, the *AS*-curve is given by Equation (1),

$$\pi = \pi^e + \omega \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + s.$$

Within this variant of the extended *AS-AD* model we can now examine the medium-run macroeconomic effects of the combination of shocks that occurred at the outbreak of the COVID-19 pandemic:

+ – an increase in the fraction of consumption expenditure and/or exports not occurring due to lockdowns and/or social distancing domestically and/or abroad, $i \uparrow / i^* \uparrow$
+ – a decrease of productivity, $z \downarrow$.

To mirror the situation of the Euro Area in early 2020, the workbook *Extended-AS-AD-Model-Pandemics.xlsm* calculates the effects of these shocks for an economy that, mirroring the situation of the Euro Area in early 2020, initially is in a deep crisis outcome (the calculations also take into account adaptive inflation expectations):

Workbook for the Extended *AS-AD* Model

The workbook *Extended-AS-AD-Model-Pandemics.xlsm* allows to study how combinations of aggregate demand and supply shocks that occur at the (renewed) outbreak of a pandemic ...

Extended AS-AD Model (Pandemics): Parameter Specification

				TR-Curve Parameters															IS-Curve Parameters												AS-Curve Parameters	
Period	z	$\bar{z}$	ϕ_z	r_0^{MP}	r_0^{RP}	r_y	ϕ_y	ϕ_π	$\bar{Y}$	π^T	ι	ι^*	C_r	I_r	TB_ε	ε_r	C_y	TB_{im}	G_y	A	C_0	I_0	G_0	TB_0	ε_0	T	TB_{ex}	Y^*	T^*	r^*	ω	s
0	1	1	1	0.020	0.050	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
1	1	1	1	0.020	0.050	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.180	1.500	2.400	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
2	0.96	1	1	0.020	0.050	0.005	0.500	2.000	15.000	0.020	0.15	0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.133	1.500	2.400	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
3	0.96	1	1	0.020	0.050	0.005	0.500	2.000	15.000	0.020	0.15	0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.133	1.500	2.400	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
4	0.96	1	1	0.020	0.050	0.005	0.500	2.000	15.000	0.020	0.15	0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.133	1.500	2.400	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
5	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	-0.15	-0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.828	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
6	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	-0.15	-0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.828	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
7	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	-0.15	-0.1	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.828	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
8	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
9	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
10	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
11	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
12	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
13	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
14	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
15	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
16	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
17	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
18	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
19	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
20	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
21	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
22	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
23	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
24	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
25	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
26	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
27	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
28	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
29	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
30	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
31	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
32	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
33	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
34	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
35	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
36	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
37	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
38	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
39	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
40	1	1	1	0.020	0.010	0.005	0.500	2.000	15.000	0.020	0	0	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000

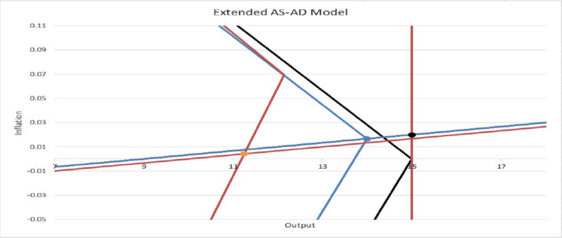
… and affect output, inflation and the other endogenous variables in the medium-run macroeconomic outcome:

Extended AS-AD Model (Pandemics): Quantitative Results and Extended AS-AD Diagram

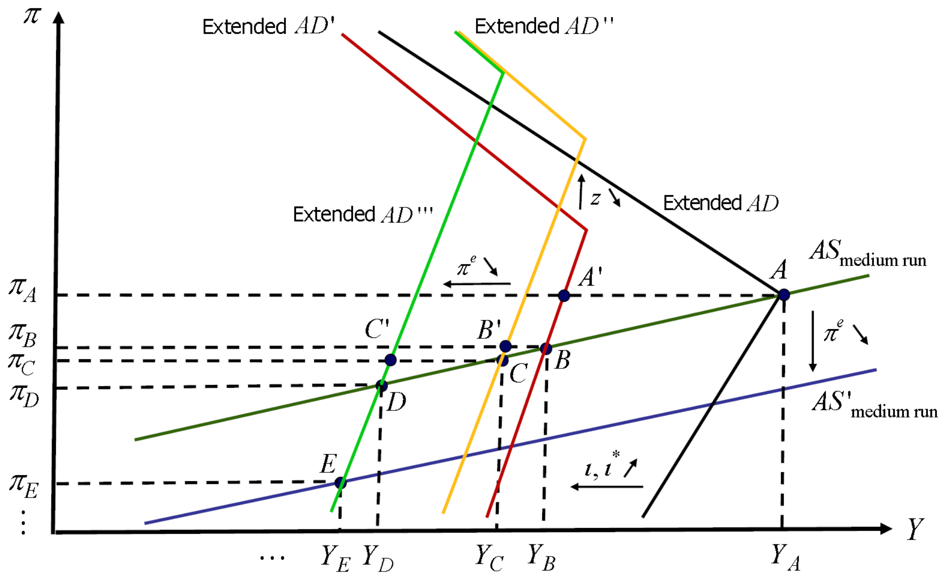
Medium-Run Macroeconomic Outcomes

	Period 0	1	2	1000
Inflation	0.020	0.017	0.004	0.020
Output	15.000	14.000	11.246	15.000
Real Interest Rate	0.070	0.030	0.055	0.030
Consumption	8.420	7.980	5.295	8.580
Investment	1.880	1.920	1.526	2.520
Government Expenditure	2.800	2.950	3.363	2.800
Trade Balance	0.860	1.150	1.070	1.100
Real Exchange Rate	0.760	1.000	0.852	1.000
Nominal Monetary Policy Rate	0.040	0.000	0.000	0.040
Nominal Interest Rate	0.090	0.047	0.059	0.050

Period — Current: 2 — Plus — Minus



Turning to a detailed graphical analysis of the type of effects depicted in the workbook:



Movement from Point A to Point B:

- Following the increases of i and i^*, the graph depicts (as in the workbook) the case of the extended AD-curve shifting to the left and upwards, and the slopes of both portions of the extended AD-curve becoming steeper.
- Following this shift of the extended AD-curve, for a given rate of inflation there is now a lower level of aggregate demand and thus output (initial drop in aggregate demand plus Keynesian multiplier effects); the central bank cannot lower the monetary policy rate as the latter already is at the zero lower bound (from Point A to Point A').
- As output falls, the wage mark-ups and subsequently inflation fall as well (adjustment towards the medium-run AS-curve). Therefore, the endogenous component of the risk premium increases, reflecting continued increases in $Y_t^{ZLB}\big|_{\pi}$. (As the zero lower bound for the monetary policy rate becomes effective at increasingly higher levels of output, the endogenous component of the risk premium keeps increasing.)
- This increase of the risk premium in turn implies an increase of the interest rate at which households and firms can borrow, and thus crowds out consumption, investment and the trade balance, until Point B is reached.

Movement from Point *B* to Point *C*:

- Following the decrease of z, the extended AD-curve shifts parallel upwards: from (49) we observe that π^{ZLB} increases, and from (48) that Y^{ZLB} is unaffected.
- From (47), $Y^{ZLB}\big|_{\pi}$ increases, and so as the zero lower bound for the monetary policy rate becomes effective at a higher level of output, the endogenous component of the risk premium rises. The implied rise of the real interest rate causes, for a given rate of inflation, crowding out of consumption, investment and the trade balance (from Point B to Point B').
- The adjustment from Point B' to Point C follows the same economic rationale as the adjustment from Point A' to Point B.

Movement from Point C to Point E:

+ This is the same type of dynamics as we had analyzed for medium–run adjustment following global financial crisis shocks (a collapse in housing prices and an increase in the exogenous component of the risk premium). In particular, under adaptive inflation expectations, given the decrease in the actual rate of inflation from Point A to Point C, inflation expectations will be revised downward in the second period of adjustment.
+ This leads to a further increase of the real interest rate for a given actual rate of inflation, causing further crowding out of consumption, investment and the trade balance (from Point C to Point C').
+ As the actual rate of inflation keeps falling in the adjustment towards the medium–run AS–curve, the endogenous component of the risk premium increases yet further, reflecting continued increases in $Y^{ZLB}\big|_{\pi}$, as per Equation (47). There is thus yet further crowding out of consumption, investment and the trade balance in the movement from Point C' to Point D.
+ Not to be neglected, the decrease of inflation expectations in the second period of adjustment also causes a downward shift of the medium–run AS–curve (for a given level of output, actual inflation decreases, as wage growth and thus the growth rate of production costs fall).
+ As the rate of inflation keeps falling in the adjustment towards the new medium–run AS–curve, the endogenous component of the risk premium increases yet further, reflecting further continued increases in $Y^{ZLB}\big|_{\pi}$, as per Equation (47).
+ There is thus yet another round of crowding out of consumption, investment and the trade balance in the movement from Point D to Point E. Point E is the medium–run outcome after the second period of adjustment.
+ Full medium–run adjustment does not end at Point E, as under adaptive inflation expectations the spiral of decreases in the actual rate of inflation causing in the following period further decreases in inflation expectations keeps operating.

The workbook Extended-AS-AD-Model-Pandemics.xlsm depicts the full mediumrun adjustment (for the full sequence of shocks specified in the workbook):